

HW 10 SOLUTIONS

Practice Problems

5.1

(a) The sample space S gives the possible values for Y_1 and Y_2 :

S	AA	AB	AC	BA	BB	BC	CA	CB	CC
(y_1, y_2)	$(2, 0)$	$(1, 1)$	$(1, 0)$	$(1, 1)$	$(0, 2)$	$(1, 0)$	$(1, 0)$	$(0, 1)$	$(0, 0)$

Since each sample point is equally likely with probability $1/9$, the joint distribution for Y_1 and Y_2 is given by

	$y_1 = 0$	$y_1 = 1$	$y_1 = 2$
$y_2 = 0$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{9}$
$y_2 = 1$	$\frac{2}{9}$	$\frac{2}{9}$	0
$y_2 = 2$	$\frac{1}{9}$	0	0

(b)

$$F(1, 0) = P(Y_1 \leq 1, Y_2 \leq 0) = P(0, 0) + P(1, 0) = \frac{1}{9} + \frac{2}{9} = \frac{3}{9} = \frac{1}{3}.$$

5.5

(a)

$$P(Y_1 \leq 1/2, Y_2 \leq 1/3) = \int_0^{1/2} \int_0^{1/3} 3y_1 dy_1 dy_2 = 0.1065.$$

(b)

$$P(Y_2 \leq Y_1/2) = \int_0^1 \int_0^{y_1/2} 3y_1 \, dy_1 \, dy_2 = 0.5.$$

5.9

(a)

Since the density must integrate to 1: $1 = \int_0^1 \int_0^{y_2} k(1 - y_2) \, dy_1 \, dy_2 = k/6$, so $k = 6$

(b)

Because $Y_1 \leq Y_2$, we split into two regions (drawing a picture is useful):

$$P(Y_1 \leq \tfrac{3}{4}, Y_2 \geq \tfrac{1}{2}) = \int_{1/2}^1 \int_{1/2}^1 6(1 - y_2) \, dy_1 \, dy_2 + \int_{1/2}^{3/4} \int_{y_1}^1 6(1 - y_2) \, dy_2 \, dy_1 = \frac{24}{64} + \frac{7}{64} = \frac{31}{64}.$$

5.11

The area of the triangular region is 1, so with a uniform distribution this is the value of the density function. Again, using geometry (drawing a picture is again useful):

(a)

$$P(Y_1 \leq \tfrac{3}{4}, Y_2 \leq \tfrac{3}{4}) = 1 - P(Y_1 > \tfrac{3}{4}) - P(Y_2 > \tfrac{3}{4}) = 1 - \left(\tfrac{1}{2}\right) \left(\tfrac{1}{2}\right) \left(\tfrac{1}{4}\right) - \left(\tfrac{1}{2}\right) \left(\tfrac{1}{4}\right) \left(\tfrac{1}{4}\right) = \frac{29}{32}.$$

(b)

$$P(Y_1 - Y_2 \geq 0) = P(Y_1 \geq Y_2)$$

The region specified in this probability statement represents 1/4 of the total region of support, so $P(Y_1 \geq Y_2) = \frac{1}{4}$.

5.17

This can be found using integration (polar coordinates are helpful). But, note that this is a bivariate uniform distribution over the unit circle (radius 1), and the probability of interest represents 50% of the support. Thus, the probability is 0.5.

5.25

(a)

The marginals are: $f_1(y_1) = e^{-y_1}$, $y_1 > 0$, and $f_2(y_2) = e^{-y_2}$, $y_2 > 0$. These are exponential densities with $\beta = 1$.

(b)

$$P(1 < Y_1 < 2.5) = P(1 < Y_2 < 2.5) = e^{-1} - e^{-2.5} = 0.2858$$

(c)

Support: $y_2 > 0$.

(d)

Conditional density: $f(y_1 | y_2) = f_1(y_1) = e^{-y_1}$, $y_1 > 0$

(e)

$f(y_2 | y_1) = f_2(y_2) = e^{-y_2}$, $y_2 > 0$.

(f)

The answers are the same.

(g)

The probabilities are the same.

5.27

(a)

$$f_1(y_1) = \int_{y_1}^1 6(1 - y_2) dy_2, = 3(1 - y_1)^2, 0 \leq y_1 \leq 1$$
$$f_2(y_2) = \int_0^{y_2} 6(1 - y_2) dy_1, = 6y_2(1 - y_2), 0 \leq y_2 \leq 1$$

(b)

$$P(Y_2 \leq \frac{1}{2} \mid Y_2 \leq \frac{3}{4}) = \frac{\int_0^{1/2} \int_0^{y_2} 6(1 - y_2) dy_1 dy_2}{\int_0^{3/4} 3(1 - y_1)^2 dy_1} = \frac{32}{63}.$$

(c)

$$f(y_1|y_2) = 1/y_2, 0 \leq y_1 \leq y_2 \leq 1.$$

(d)

$$f(y_2|y_1) = 2(1 - y_2)/(1 - y_1)^2, 0 \leq y_1 \leq y_2 \leq 1.$$

(e)

From part (d): $f(y_2|1/2) = 8(1 - y_2), 1/2 \leq y_2 \leq 1$. Thus $P(Y_2 \geq 3/4|Y_1 = 1/2) = 1/4$

5.33

Refer to Ex 5.15:

(a)

$$f_1(y_1) = \int_0^{y_1} e^{-(y_1)} dy_2 = y_1 e^{-y_1}, y_1 \geq 0.$$
$$f_2(y_2) = \int_{y_2}^{\infty} e^{-(y_1)} dy_1 = e^{-y_2}, y_2 \geq 0.$$

(b)

$$f(y_1 | y_2) = e^{-(y_1 - y_2)}, \quad y_1 \geq y_2.$$

(c)

$$f(y_2 | y_1) = 1/y_1, \quad 0 \leq y_2 \leq y_1.$$

(d)

The density functions are different.

(e)

The marginal and conditional probabilities can differ.

Submitted Problems

5.2

a. The sample space for the toss of three balanced coins w/ probabilities are below:

Outcome	<i>HHH</i>	<i>HHT</i>	<i>HTH</i>	<i>HTT</i>	<i>THH</i>	<i>THT</i>	<i>TTH</i>	<i>TTT</i>
(y_1, y_2)	(3, 1)	(3, 1)	(2, 1)	(1, 1)	(2, 2)	(1, 2)	(1, 3)	(0, -1)
probabil- ity	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8

	$y_1 = 0$	1	2	3
$y_2 = -1$	$\frac{1}{8}$	0	0	0
0	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$
2	0	$\frac{1}{8}$	$\frac{1}{8}$	0
3	0	$\frac{1}{8}$	0	0

b.

$$F(2, 1) = p(0, -1) + p(1, 1) + p(2, 1) = \frac{1}{2}.$$

5.8

a. Since the density must integrate to 1, evaluate

$$\int_0^1 \int_0^1 k y_1 y_2 dy_1 dy_2 = k/4 = 1,$$

so $k = 4$.

b.

$$F(y_1, y_2) = P(Y_1 \leq y_1, Y_2 \leq y_2) = 4 \int_0^{y_2} \int_0^{y_1} t_1 t_2 dt_1 dt_2 = y_1^2 y_2^2, \quad 0 \leq y_1 \leq 1, \quad 0 \leq y_2 \leq 1.$$

c.

$$P(Y_1 \leq 1/2, Y_2 \leq 3/4) = (1/2)^2 (3/4)^2 = 9/64.$$

5.14

a. Since $f(y_1, y_2) \geq 0$, simply show

$$\int_0^1 \int_{y_1}^{2-y_1} 6y_1^2 y_2 dy_2 dy_1 = 1.$$

b.

$$P(Y_1 + Y_2 < 1) = P(Y_2 < 1 - Y_1) = \int_0^{0.5} \int_{y_1}^{1-y_1} 6y_1^2 y_2 dy_2 dy_1 = \frac{1}{16}.$$

5.16

a.

$$P(Y_1 < 1/2, Y_2 > 1/4) = \int_{1/4}^1 \int_0^{1/2} (y_1 + y_2) dy_1 dy_2 = \frac{21}{64} \approx 0.328125.$$

b.

$$P(Y_1 + Y_2 \leq 1) = P(Y_1 \leq 1 - Y_2) = \int_0^1 \int_0^{1-y_2} (y_1 + y_2) dy_1 dy_2 = \frac{1}{3}.$$

5.22

a. The marginal distributions for Y_1 and Y_2 are given in the margins of the table.

b.

$$P(Y_2 = 0 \mid Y_1 = 0) = \frac{.38}{.76} = .5 \quad P(Y_2 = 1 \mid Y_1 = 0) = \frac{.14}{.76} = .18$$
$$P(Y_2 = 2 \mid Y_1 = 0) = \frac{.24}{.76} = .32$$

c.

The desired probability is $P(Y_1 = 0 \mid Y_2 = 0) = \frac{.38}{.55} = .69$.

5.32

a.

$$f_1(y_1) = \int_{y_1}^{2-y_1} 6y_1^2 y_2 \, dy_2 = 12y_1^2(1 - y_1), \quad 0 \leq y_1 \leq 1.$$

b. This marginal density must be constructed in two parts:

$$f_2(y_2) = \begin{cases} \int_0^{y_2} 6y_1^2 y_2 \, dy_1 = 2y_2^4, & 0 \leq y_2 \leq 1, \\ \int_0^{2-y_2} 6y_1^2 y_2 \, dy_1 = 2y_2(2 - y_2)^3, & 1 \leq y_2 \leq 2. \end{cases}$$

c.

$$f(y_2 \mid y_1) = \frac{1}{2} y_2 / (1 - y_1), \quad y_1 \leq y_2 \leq 2 - y_1.$$

d. Using the density found in part c,

$$P(Y_2 < 1.1 \mid Y_1 = 0.6) = \frac{1}{2} \int_{0.6}^{1.1} \frac{y_2}{0.4} \, dy_2 \approx 0.53.$$

5.34

a. Given $Y_1 = y_1$, Y_2 has a uniform distribution on the interval $(0, y_1)$.

b. Since $f_1(y_1) = 1$, $0 \leq y_1 \leq 1$,

$$f(y_1, y_2) = f(y_2 \mid y_1) f_1(y_1) = \frac{1}{y_1}, \quad 0 \leq y_2 \leq y_1 \leq 1.$$

c.

$$f_2(y_2) = \int_{y_2}^1 \frac{1}{y_1} \, dy_1 = -\ln(y_2), \quad 0 \leq y_2 \leq 1.$$